

Benchmarking Image Aesthetic Assessment from Scoring to Reasoning

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SUMMARY

Image Aesthetic Assessment (IAA) aims to evaluate the aesthetic quality of images and model human judgments of visual appeal. Recent advances in multimodal models have expanded IAA beyond scalar score prediction to language-based evaluation and explanatory analysis. We interpret this expansion as a paradigm transition from aesthetic score-centric prediction IAA to aesthetic reasoning-oriented IAA, where models are increasingly expected to evaluate aesthetic quality and justify their judgments with visual evidence and aesthetic criteria. In this paper, we present a comprehensive review of IAA methods, datasets, and evaluation protocols, together with a unified empirical benchmark for representative models on score prediction and language-based aesthetic assessment. We investigate the paradigm transition by organizing existing methods into three categories, aesthetic score prediction based on aesthetic features, aesthetic assessment based on vision-language models (VLMs), and aesthetic reasoning based on multimodal large language models (MLLMs). To quantitatively depict this transition, we conduct extensive experiments and construct comprehensive benchmarks by evaluating diverse IAA models, including score-centric vision models, General MLLMs, and IAA-specific MLLMs on public datasets, AVA, TAD66K, and UNIAA-Bench.

IAA score prediction benchmarks

IAA reasoning benchmarks

Lexical and Semantic Evaluations VQA Evaluations

We demonstrate that score-centric vision models still dominate the IAA score prediction tasks, whereas MLLM-based models provide stronger support for multi-dimensional and in-depth aesthetic reasoning and analysis from a professional perspective. We further reveal that as the model size increases, the consistency trend between IAA performance and human preference ranking improves in score prediction, which to some extent validates the scaling law of large language models. We also observe current IAA models show limited cross-dataset generalization due to their sensitivity to dataset-specific rating distributions and annotation preferences. For language-based aesthetic assessment, we find that no single metric fully captures aesthetic reasoning quality: General MLLMs show comparable surface-level text matching to IAA-specific MLLMs, whereas the strongest IAA-specific MLLMs show clearer advantages in semantic alignment and VQA (Visual Question Answering)-style perception.

General MLLMs, and IAA-specific MLLMs

We finally release a unified IAA evaluation pipeline for unified score prediction and language-based aesthetic assessment, which are available at <https://github.com/An-Moon/Aesthetic-evaluation>

Beside providing a comprehensive and quantifiable IAA benchmark, this study also have the

potential to shape the future of machine intelligence with cognitively and culturally sense of aesthetic. 46 47

KEYWORDS 48

Image Aesthetic Assessment, aesthetic scoring, aesthetic reasoning, vision-language models, 49 multimodal large language models 50

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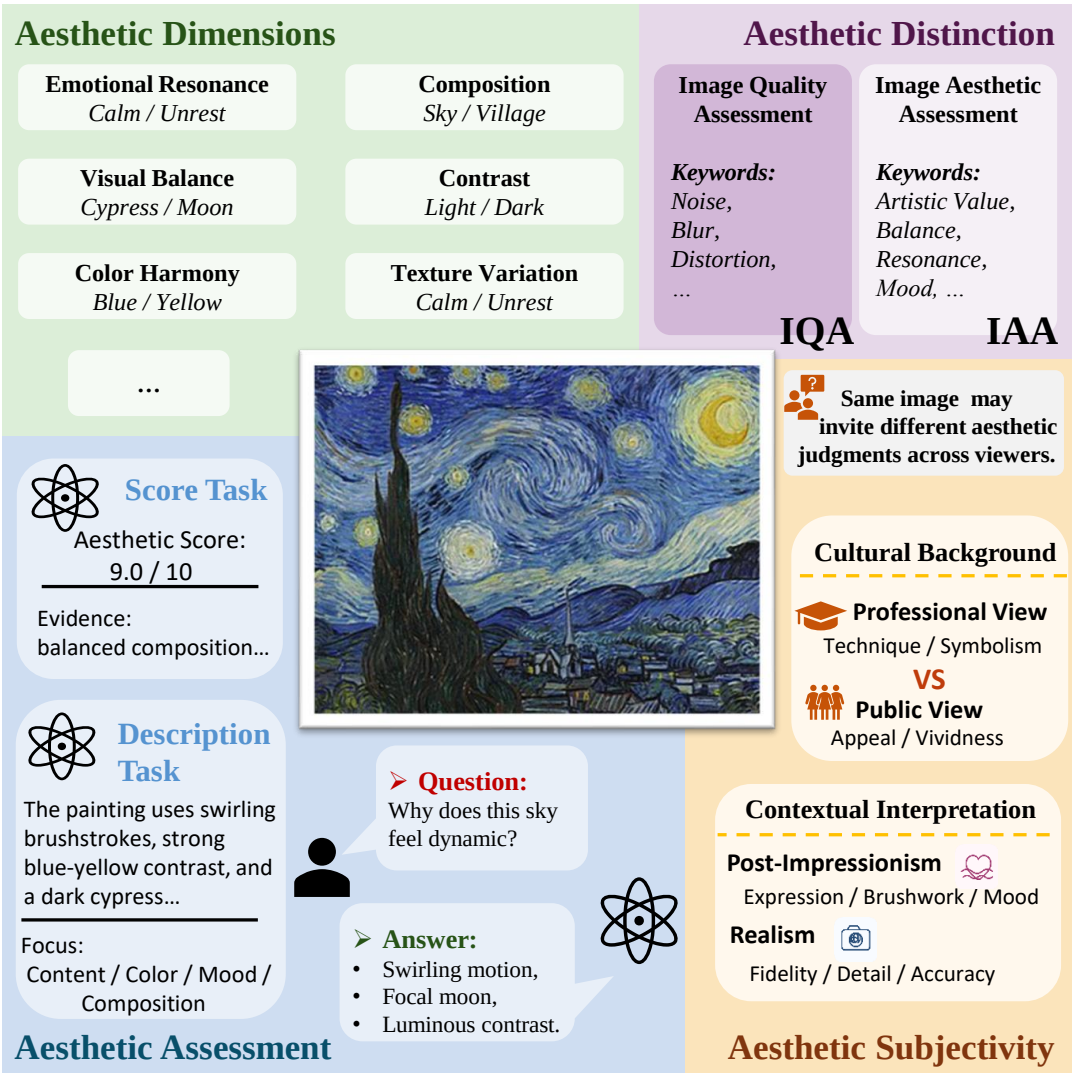


Figure 1: An intuitive overview of Image Aesthetic Assessment (IAA) using Van Gogh’s *The Starry Night* as an example. The figure illustrates aesthetic dimensions, the distinction between IQA and IAA, representative task forms such as score prediction, description generation, and VQA-style perception, as well as the subjectivity across viewers. 52

As French artist Edgar Degas says “Art is not what you see, but what you make others see”. 52 However, making rigid artificial intelligence (AI) see the aesthetic from vivid and professional per- 53 spective is a huge challenge, but it is also a necessary path for machines to move towards gener- 54 al intelligence. Recent study also showcases that current advanced AI, e.g., vision-language 55 models trend to converge on overly generic epistemic outputs¹ in cultural and aesthetic domains, 56

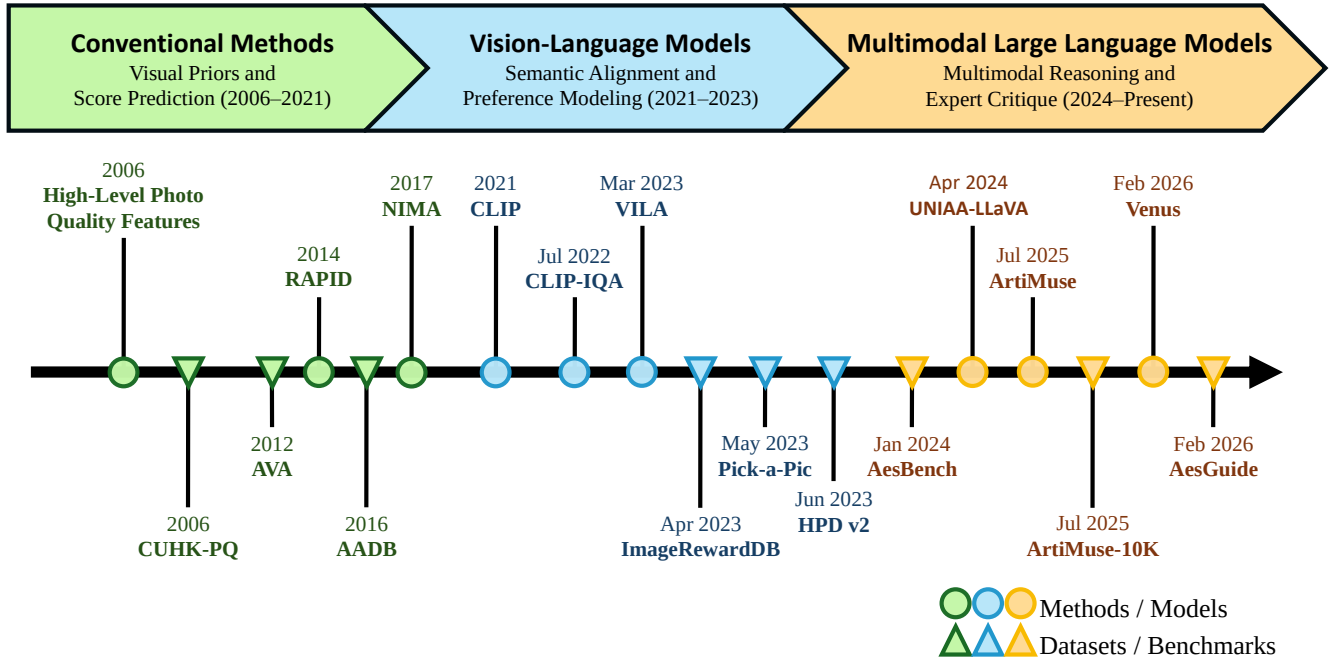


Figure 2: Evolution timeline of Image Aesthetic Assessment (IAA). The timeline shows the progression from conventional score prediction to vision-language alignment and MLLM-based aesthetic reasoning.

and cause the concern about AI epistemic loss phenomenon². This new trend has created a growing demand for AI models that can understand high-level visual semantics and aesthetic quality. Image Aesthetic Assessment (IAA) addresses this need by predicting aesthetic quality and modeling human aesthetic judgments. Unlike Image Quality Assessment (IQA), which focuses on low-level degradations such as noise and blur, IAA focuses on higher-level attributes such as composition, artistic value, and emotional response³. Fig. 1 further illustrates that IAA evaluates images through multiple aesthetic perspectives, such as composition, emotional response, visual balance, and viewer-dependent interpretation, and can support language-based reasoning that explains why an image is perceived as aesthetically effective.

Due to the subjective and context-dependent nature of aesthetics, IAA remains a challenging problem. Specifically, aesthetic perception varies across individuals and cultural groups, influenced by their background and expertise. For example, general viewers may prefer rich colors and direct visual appeal, whereas professional photographers tend to emphasize lighting, composition, and technical quality. Therefore, both expert and crowdsourced annotations can exhibit biases. In addition, aesthetic evaluation is highly dependent on the context, with evaluation criteria varying across domains. For instance, photography often emphasizes realism, whereas abstract art may emphasize symbolism. These characteristics introduce uncertainty into aesthetic modeling and make reliable annotation difficult.

To address these challenges, IAA methods have evolved across different modeling paradigms. Early methods relied on handcrafted features and later convolutional neural networks (CNNs) to predict aesthetic scores from visual inputs^{4,5}. With the rise of vision-language models (VLMs), IAA began to connect visual aesthetic cues with textual concepts through image-text alignment and textual supervision^{6,7}. More recently, multimodal large language models (MLLMs) have extended IAA toward language-based aesthetic judgment and explanation, enabling models to generate descriptive and structured outputs beyond scalar scores⁸. As illustrated in Fig. 2, this evolution represents a transition from feature-based score modeling to language-based aesthetic analysis. Table 1 summarizes this methodological evolution and the representative literature in

each stage.

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Table 1: Taxonomy of Image Aesthetic Assessment (IAA) based on methodological evolution. The table organizes representative methods into three stages: Conventional IAA, Vision-Language Alignment, and Generative Aesthetic Reasoning.

Era & Paradigm	Core Methodology	Representative Literature
Conventional IAA (Pre-2021)	Hand-crafted Features	Datta <i>et al.</i> ⁹ , Ke <i>et al.</i> ¹⁰ , Joshi <i>et al.</i> ¹¹
	Photographic Rules	Luo <i>et al.</i> ¹² , Mai <i>et al.</i> ¹³
	Deep Representations	Lu <i>et al.</i> ⁴ , NIMA ¹⁴ , MUSIQ ¹⁵ , AFDC ¹⁶
Vision-Language Alignment (2021-2023)	Contrastive Zero-shot	CLIP ⁶ , CLIP-IQA ¹⁷ , LAION-5B ¹⁸
	Text-augmented Learning	LIQE ¹⁹ , UniQA ²⁰ , VILA ²¹ , BLIP ²²
	Preference Alignment	ImageReward ²³ , HPS v2 ²⁴ , Pick-a-pic ²⁵
Generative Aesthetic Reasoning (2023-Present)	General MLLM Foundations	LLaVA ²⁶ , InstructBLIP ²⁷ , MiniGPT-4 ²⁸
	Aesthetic Expert MLLMs	UNIAA ²⁹ , AesExpert ³⁰ , ArtiMuse ³¹ , PGT-IAQA ³²
	Bias & Personalization	AesBench ⁸ , AesBiasBench ³³ , Zhong <i>et al.</i> ³⁴

Several surveys have reviewed the development of IAA from different perspectives. Deng et al.³ conducted an early experimental survey on aesthetic modeling with hand-crafted and deep features, mainly focusing on score prediction. Yang et al.³⁵ extended this line of work by summarizing datasets, task types, and evaluation methods. Anwar et al.³⁶ reviewed traditional and deep learning-based IAA methods and compared quantitative results across public benchmarks, highlighting major challenges and possible solutions. Daryanavard Chouchenani et al.³⁷ recently published a decade-long review of deep learning-based IAQA. They organized the literature by input, process, and output stages and analyzed issues such as dataset limitations and environmental factors. **However, these surveys do not fully examine the recent change in task formulation: IAA is no longer defined only by score prediction, but increasingly involves language-based judgment and explanation. We use this scoring-to-reasoning transition as the organizing perspective of this survey and benchmark.**

Based on this perspective, this paper makes four contributions: (1) We propose a unified taxonomy that organizes IAA methods into three stages: score-centric visual modeling, VLM-based semantic alignment, and MLLM-based aesthetic reasoning. (2) We analyze the evolution of IAA datasets and metrics, showing how the field has expanded from scalar ratings to language-based supervision and perception-oriented reasoning. (3) We construct a unified empirical benchmark covering score prediction, description generation, and VQA-style perception on AVA, TAD66K, and UNIAA-Bench. (4) We provide systematic findings on model family differences, model scale, cross-dataset generalization, and language-based aesthetic reasoning, and release a unified evaluation pipeline to support reproducible IAA evaluation.

As summarized in Fig. 3, the remainder of this paper follows this scoring-to-reasoning narrative. We first review feature-based aesthetic modeling methods, then discuss VLM-based semantic alignment and MLLM-based aesthetic reasoning. We next introduce datasets and evaluation metrics, present the empirical benchmark, and conclude with a discussion of future directions.

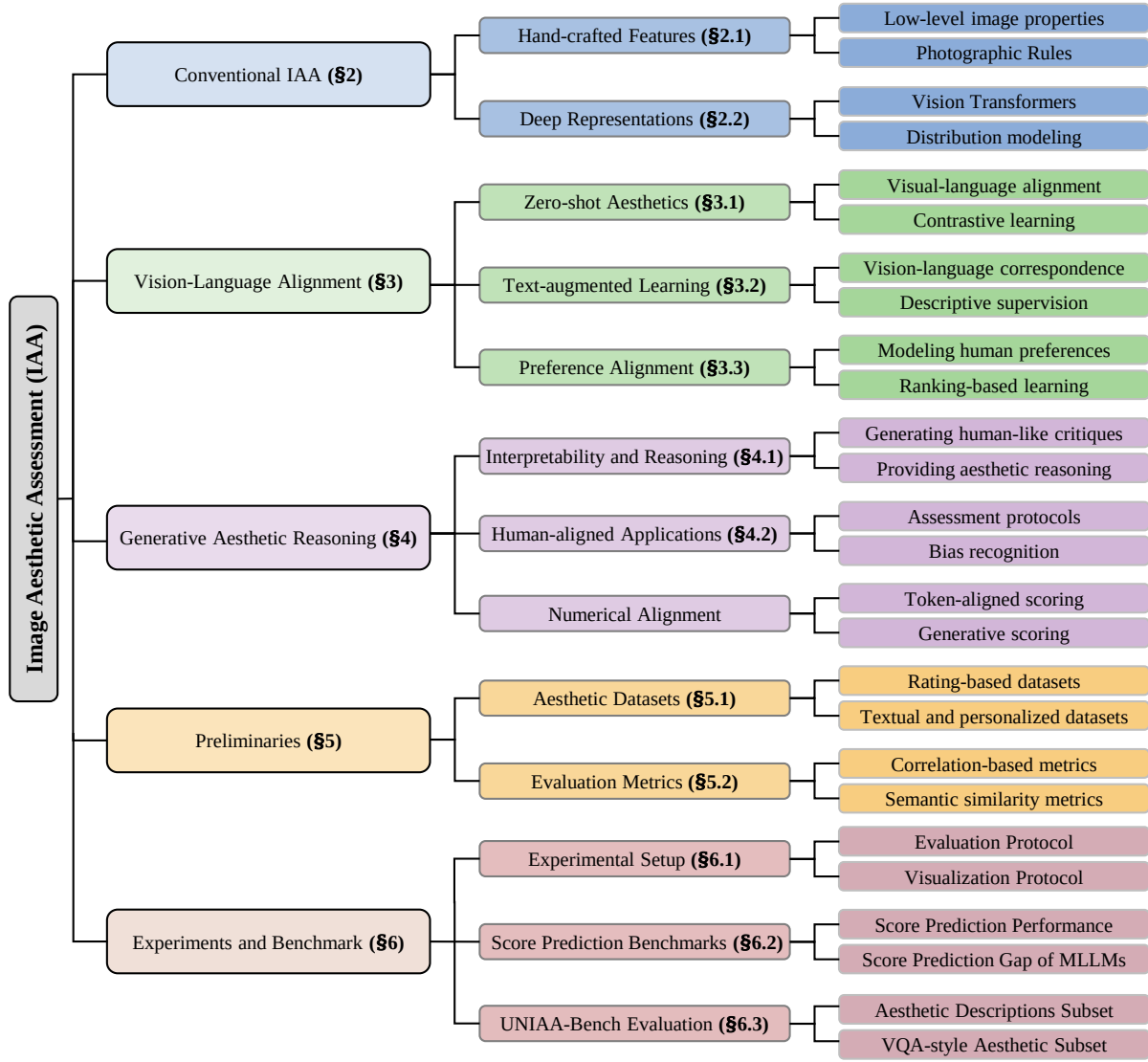


Figure 3: Overview of the paper structure. The figure summarizes the major sections, subsections, and core themes covered in this paper.

Aesthetic Scoring based on Conventional Methods

This section reviews the score-prediction foundation of IAA. In this stage, most methods formulate aesthetic assessment as learning a mapping from visual features to human aesthetic scores. Traditional methods can be broadly divided into two categories: handcrafted methods and deep learning methods. Handcrafted methods rely on predefined aesthetic features, whereas deep methods learn representations directly from data. Although these approaches were developed in the earlier stages, they remain important for understanding IAA development. They provide insights into how aesthetic features can be effectively modeled. Understanding these approaches is important for analyzing the evolution of modern IAA models. Table 2 summarizes representative conventional IAA methods and compares their use of hand-crafted features, deep CNNs, Transformers, and distribution modeling.

Hand-crafted Features

A central challenge in image aesthetic assessment is bridging the gap between low-level visual features and high-level human perception. Early studies suggest that visual attributes,

such as color harmony, contrast, and texture, play an important role in aesthetic perception⁹. These attributes capture fundamental visual cues and provide the basis for higher-level aesthetic judgments. Based on this observation, early approaches modeled aesthetic preferences by extracting measurable visual features from images. Subsequent efforts have extended these representations by introducing handcrafted descriptors and affective attributes to capture emotional responses^{38–42}. These methods have been systematically summarized in prior survey works^{11,36}.

Table 2: Feature comparison of representative conventional IAA methods. The table compares whether each method uses hand-crafted features, deep CNNs, Transformers, or distribution modeling, providing a compact overview of conventional score prediction approaches.

Method	Hand-crafted	Deep CNN	Transformer	Distribution
Datta <i>et al.</i> ⁹	✓	×	×	×
Ke <i>et al.</i> ¹⁰	✓	×	×	×
RAPID ⁴	×	✓	×	×
NIMA ¹⁴	×	✓	×	✓
A-Lamp ⁵	×	✓	×	×
MUSIQ ¹⁵	×	×	✓	✓
AesMamba ⁴³	×	×	×	✓

Researchers have moved beyond basic statistical features by formalizing photographic rules into computational metrics. Classic works translated qualitative concepts in the photography profession, such as the rule of thirds, visual balance, and depth of field, into quantitative metrics¹⁰. These methods evaluate image aesthetic metrics by segmenting regions that are aesthetically salient and extracting aesthetic features¹². Although these rule-based representations are limited in their flexibility, they provide important structural priors, reflecting the assumption that aesthetic quality is closely related to spatial composition patterns. These priors remain valuable for modeling the image composition and aesthetic structure. They have been widely applied in tasks such as aesthetic-driven image cropping^{44–46}, image enhancement^{47,48}, and style transfer⁴⁹.

Deep Learning-based Methods

Deep learning has shifted IAA from manually designed feature pipelines to end-to-end representation learning. Early methods primarily relied on convolutional neural networks (CNNs) to learn aesthetic features directly from images^{50,51}. To capture more complex aesthetic factors, subsequent studies explored multi-task and multi-branch architectures to model aesthetic scores and semantic information^{52,53}, while others used dual-channel or multi-stream networks to capture global composition and local details separately¹³. Furthermore, research has enhanced representation capabilities through improved structural design⁵⁴. The introduction of residual networks allows for deeper and more stable model training, resulting in stronger feature representation capabilities. To address the subjectivity in aesthetic judgments, some studies have modeled aesthetic scoring as a probability distribution^{55,56}. Graph-based methods have been introduced to explicitly model the relationships between image regions, thus providing a structured representation of the visual composition⁵⁷. Later, Transformer-based architectures were adopted, introducing self-attention mechanisms to effectively encode visual elements and aesthetic features. Vision Transformers and their hierarchical variants enable global context modeling, further improving the representation of high-level aesthetic semantics^{43,58–60}. Overall, deep learning has shifted IAA from shallow feature extraction to hierarchical and structured representation learning, enabling more accurate and expressive aesthetic modeling.

Table 3: Comparison of representative VLM-based IAA methods. The table summarizes whether each method supports zero-shot prediction, text supervision, preference learning, or generative outputs, providing an overview of vision-language alignment approaches in IAA.

Method	Zero-shot	Text Supervision	Preference	Generative
CLIP ⁶	✓	✓	×	×
CLIP-IQA ¹⁷	✓	✓	×	×
VILA ²¹	×	✓	×	×
LIQE ¹⁹	×	✓	×	×
BLIP ²²	×	✓	×	✓
ImageReward ²³	×	✓	✓	×
HPS v2 ²⁴	×	✓	✓	×

Vision-language alignment serves as an intermediate stage between feature-based aesthetic scoring and MLLM-based aesthetic reasoning. It introduces image-text alignment, semantic supervision, and preference modeling into IAA, allowing aesthetic quality to be connected with textual concepts and human preferences. This section reviews how IAA moves from score prediction toward cross-modal semantic alignment. Table 3 summarizes representative VLM-based IAA methods and compares their support for zero-shot prediction, text supervision, preference learning, and generative outputs.

Zero-shot Aesthetics

Contrastive Language-Image Pre-training (CLIP)⁶ introduced a new paradigm based on contrastive vision-language learning. By training dual-encoder architectures on large-scale image-text pairs, these models map visual features and textual concepts into a shared embedding space. Building on this framework, CLIP-IQA¹⁷ explores zero-shot aesthetic assessment by computing the cosine similarity between image embeddings and text prompts, such as “good photo” and “bad photo.” This result shows that aesthetic-related concepts can be expressed through simple textual descriptions. Beyond zero-shot evaluation, the learned embedding space also supports linear probing. For example, the LAION-Aesthetics predictor¹⁸ trains a lightweight classifier on frozen CLIP features to identify high-quality images from large-scale web data. This approach suggests that aesthetic information can be effectively captured in contrastive embedding spaces and used for large-scale aesthetic scoring and ranking.

Text-augmented Learning

Text-augmented learning extends VLM-based IAA by using textual supervision for aesthetic assessment. However, general vision-language alignment models often lack detailed semantic descriptions for fine-grained aesthetic analysis. Consequently, studies adapt vision-language models to image quality and aesthetic tasks through captions, professional critiques, and aesthetic attribute descriptions. Representative methods such as LIQE¹⁹ frame image assessment as a vision-language correspondence problem, extracting semantics from textual descriptions of distortions and aesthetic attributes. Similarly, UniQA²⁰ established a unified framework for quality and aesthetics, learning unified representations that combine structural and semantic information. To improve semantic understanding, recent approaches have incorporated textual annotations, such as captions and professional critiques, as supervision signals. Datasets such

as PCCD⁶¹ and AVA-Captions⁶² enable models to learn aesthetic-related semantics beyond low-level visual features. Models such as VILA²¹ operationalize this setting by leveraging image-text supervision to improve descriptive aesthetic representations.

Preference Alignment

Preference alignment further extends vision-language learning by explicitly modeling human aesthetic preferences. With the development of multimodal generative models, aesthetic assessment is no longer limited to score prediction and aims to align model outputs with human judgments. Recent studies have shown that simple contrastive alignment is insufficient to capture the subjective nature of aesthetic preference. To address this, methods such as ImageReward²³ and Pick-a-pic²⁵ introduced large-scale human preference datasets and adopted ranking-based learning to model relative aesthetic quality.

Building on this idea, approaches such as the Human Preference Score (HPS)²⁴ further fine-tune vision-language models using human feedback, improving their ability to reflect subjective aesthetic judgments. Compared with traditional regression-based methods, preference learning provides a more flexible and human-centered formulation for aesthetic assessment. This shift marks a transition from score prediction to preference modeling, laying the foundation for reasoning-based aesthetic assessments using multimodal large models.

Overall, vision-language alignment has evolved from similarity-based zero-shot evaluation to supervised semantic learning and human preference modeling. However, many representative VLM-based methods still express aesthetic assessment through similarity scores, preference rankings, or lightweight prediction heads. Although these methods introduce semantic supervision and preference signals, they are less suited to generating structured aesthetic explanations. This limitation motivates the transition toward MLLM-based aesthetic reasoning, which we discuss in the next section.

Aesthetic Reasoning based on Multimodal Large Language Models

Following the transition from feature-based scoring to semantic alignment, MLLMs further change the task form of IAA. They allow aesthetic assessment to be regarded as a conditional reasoning process, where images and textual queries jointly guide the generation of scores, choices, attributes, and explanations. Table 4 summarizes representative methods by their support for scoring, critique generation, open-ended VQA, and bias awareness. This comparison shows how MLLMs broaden the task scope of IAA from numerical assessment to interpretable and human-oriented analysis.

Interpretability and Reasoning

IAA is transitioning toward reasoning-based evaluation with the development of MLLMs, as represented by general multimodal systems such as LLaVA²⁶ and instruction-following architectures such as InstructBLIP²⁷. Earlier IAA methods mainly mapped visual features to aesthetic scores^{9,14} or aligned images with text through VLMs⁶. In contrast, recent IAA-oriented MLLMs, such as UNIAA²⁹ and ArtiMuse³¹, generate structured aesthetic analyses in natural language. Under this paradigm, aesthetic assessment can be viewed as a hierarchical process of perception → description → reasoning → judgment, which better reflects human aesthetic cognition and reasoning processes.

Table 4: A multidimensional capability comparison is presented for representative MLLM-based IAA methods. The table summarizes whether each method supports scoring, critique, open-ended VQA, and bias awareness, thus providing an overview of reasoning-oriented IAA methods.

Method	Scoring	Critique	Open-ended VQA	Bias Awareness
LLaVA ²⁶	×	✓	✓	×
Q-Bench ⁶³	✓	×	✓	×
UNIAA ²⁹	✓	✓	✓	×
AesExpert ³⁰	✓	✓	✓	×
ArtiMuse ³¹	✓	✓	✓	×
AesBiasBench ³³	✓	✓	✓	✓
ROC4MLLM ⁶⁴	✓	×	×	×

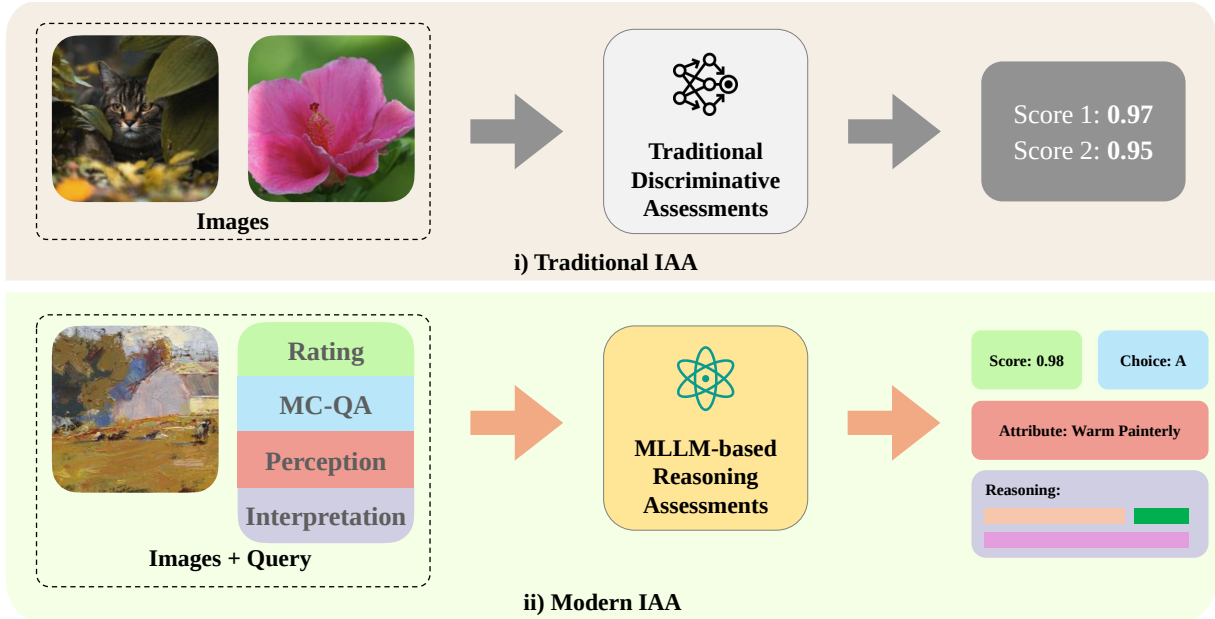


Figure 4: Workflow comparison between traditional (i) and modern (ii) paradigms in Image Aesthetic Assessment (IAA). Traditional discriminative assessment typically takes images as input and outputs scalar aesthetic scores. In contrast, modern MLLM-based reasoning assessment can be conditioned on both images and textual queries, producing richer outputs such as aesthetic scores, multiple-choice judgments, and aesthetic attributes and explanations.

To further illustrate this paradigm, Fig. 4 contrasts the typical workflows of the traditional and modern IAA paradigms. Traditional methods follow a discriminative pipeline that maps images directly to scalar aesthetic scores. In contrast, MLLM-based approaches adopt a task-conditioned framework, taking images with textual queries and producing diverse outputs, including scores, structured judgments, and descriptive analyses. This difference reflects the broader shift of IAA from black-box regression to language-mediated aesthetic reasoning.

Several recent works have explored the application of MLLMs in aesthetic assessment. UNIAA²⁹ introduced an instruction-based framework that adapts existing aesthetic datasets into instruction-following formats, enabling MLLMs to perform aesthetic reasoning tasks. AesExpert³⁰ further incorporates large-scale human feedback, improving the model’s ability to generate detailed and consistent aesthetic explanations. ArtiMuse³¹ proposes a large expert annotation dataset, which combines aesthetic scoring with language understanding, and designs a special token-level numerical prediction mechanism to compensate for the shortcomings of pure generative models in quantitative evaluation. These methods represent different strategies for integrating MLLMs into IAA, including instruction tuning, human feedback learning, and the joint modeling of scoring and

reasoning.

Other studies have focused on data-centric and text-assisted strategies. For example, PGT-IAQA³² leverages generated textual feedback to enhance aesthetic representation. ArtQuant⁶⁵ further constructs long-form aesthetic descriptions and hierarchical aesthetic elements through automatic data construction, while integrating high-precision score distribution estimation to improve alignment with human aesthetic ratings.

Overall, these methods demonstrate a consistent shift from score-based prediction to reasoning-driven aesthetic assessments. By integrating visual perception, language generation, and structured reasoning, MLLM-based approaches enable more comprehensive and interpretable aesthetic analyses.

Human-aligned Applications

The discussion above shows that IAA has moved from score prediction to reasoning-based analysis. The next step is to explore how these reasoning capabilities can be applied to practical aesthetic evaluation tasks. Recent studies have discussed this direction from multiple perspectives, including evaluation protocols, bias analysis, preference modeling, and numerical alignment⁶⁶.

The evaluation of reasoning-based outputs remains a key challenge. Traditional metrics, such as PLCC and SRCC, are designed for score prediction and cannot assess the quality of explanations. To address this limitation, AesBench⁸ introduced an expert-driven benchmark that evaluates models across multiple dimensions, including perception, empathy, assessment, and interpretation. This provides a more structured framework for assessing reasoning ability in aesthetic tasks.

Simultaneously, bias and subjectivity become more prominent in human-centered settings. AesBiasBench³³ systematically analyzes demographic bias in MLLMs using metrics such as Identity Frequency Disparity (IFD) and Normalized Representation Disparity (NRD). The results indicate that model outputs can be sensitive to demographic attributes, highlighting the need for bias-aware evaluation in aesthetic assessment. This concern is also relevant to generative visual systems, where recent work has shown that text-to-image models require value-alignment mechanisms to reduce toxic or biased outputs⁶⁷.

Another important direction is preference modeling. Personalized Image Aesthetic Assessment (PIAA) frameworks³⁴ incorporate user feedback to adapt aesthetic judgments over time. In addition, datasets such as HumanBeauty⁶⁸ combine language-based reasoning with score prediction, enabling models to better capture individual aesthetic preferences of the users. These approaches improve the flexibility of aesthetic assessment and support a more user-aligned evaluation.

Additionally, recent studies have investigated how MLLMs can generate reliable numerical scores. Studies such as Next Token Is Enough⁶⁹ have shown that aesthetic scoring can be integrated into a generative framework by modeling scores as token sequences. Methods such as ROC4MLLM⁶⁴ further improve performance by separating numerical and textual representations during training. These findings suggest that numerical alignment depends more on representation design and supervision strategies than on model architecture. Overall, MLLM-based methods extend IAA beyond score prediction toward language-based aesthetic assessment.

Datasets and Evaluation Metrics

The scoring-to-reasoning shift is also reflected in how IAA datasets and metrics are designed. As tasks move from numerical prediction to language-based judgment, datasets must provide richer

supervision, and metrics must evaluate not only score agreement but also semantic relevance and explanatory quality.

Aesthetic Datasets

Unlike conventional visual recognition tasks, IAA is fundamentally shaped by how aesthetic judgments are collected and represented. This dependence on datasets is especially important in IAA, because rating distributions, annotation granularity, and textual supervision determine whether a benchmark supports score regression, attribute-level analysis, or language-based aesthetic assessment. This view is consistent with broader discussions on machine learning datasets, where datasets are not only training resources but also task definitions and evaluation instruments⁷⁰. Accordingly, this section reviews the evolution of aesthetic datasets and their underlying annotation paradigms.

The rating-centric paradigm operates on the premise that aesthetic quality can be quantified. In the early stages of research, owing to the scarcity of large-scale data, this task was simplified into a binary classification problem. The representative dataset CUHK-PQ^{10,12} standardized this approach by categorizing over 17,000 images into distinct quality groups. A key design of CUHK-PQ is its explicit separation of professional photography from amateur snapshots, forcing the models to learn the visual gap between the two. However, treating beauty as a binary label is inherently limited because it fails to distinguish between a masterpiece and an image that is merely acceptable. To capture the continuous nature of aesthetics, the field has shifted towards score regression. The release of AVA⁷¹ marked a major shift. Unlike previous datasets that offered limited scales, AVA provides a 1-to-10 rating derived from approximately 200 voters per image. The critical innovation of AVA is that it preserves the full score distribution rather than just a single mean score. This allows models to capture the diversity of human opinion, recording not only the consensus but also the disagreement. Despite its success, a single numerical score acts as a black box: it indicates how good an image is but not why. To address this interpretability gap, AADB⁷² introduced attribute annotations for images. Beyond the global score, it annotates specific photographic factors, such as the rule of thirds and color harmony, marking a step closer to semantic understanding.

While attribute annotations allow for fine-grained analysis, they have limited capacity to express complex aesthetic semantics: a fixed vocabulary of discrete tags cannot articulate the complex syntactic relationships inherent in aesthetic perception. To address this limitation, the field has shifted towards a text-augmented paradigm, extending aesthetic modeling toward vision-language understanding. The PCCD dataset⁶¹ represents the first major attempt to capture subjective critiques of aesthetics. By collecting detailed sentence-level comments on composition and lighting, the PCCD moves beyond binary labels to model the reasoning behind aesthetic judgment. Conversely, to facilitate large-scale pre-training, recent studies have focused on descriptive captions. AVA-Captions⁶² scales up textual supervision by associating more than 230,000 images with descriptive sentences. Unlike subjective critiques, these captions provide a grounded, neutral description of visual content, serving as a critical bridge to align aesthetic features with the semantic space of modern MLLMs.

Conventional aesthetic paradigms operate under the reductive assumption of universal consensus, treating the variance in human ratings as statistical noise to be suppressed. However, the personalization-oriented paradigm challenges this view, positing that aesthetic appreciation is inherently subjective and that opinion divergence reflects underlying user diversity. Consequently, the research focus has shifted from fitting a single ground-truth distribution to modeling the mapping between visual content and individual preferences. To operationalize this shift, specialized datasets have been developed. The Flickr-AES dataset⁷³ explicitly encodes the discrep-



Figure 5: Examples of representative IAA benchmark contents. The figure shows score-level ratings, VQA-style aesthetic perception questions, and language-based reasoning annotations, demonstrating the diverse supervision formats used in current IAA benchmarks.

ancy between professional photographers, who prioritize technical attributes, and the general public, who are primarily driven by content semantics. Furthermore, recent advancements have transcended simple demographic labels by integrating rich semantic contexts and user profiles. The PARA dataset⁷⁴ represents a significant leap, annotating images across specific scenarios while simultaneously recording the annotators’ psychological traits, such as Big-Five personality priors. By incorporating these priors, modern architectures can predict aesthetic judgments conditioned on the viewer’s intrinsic character. Recent advancements also explicitly address the challenge of subjective personalization by incorporating personality priors and active learning frameworks^{75,76}.

Table 5: Applicability of evaluation metrics across different IAA paradigms. Representative metrics are drawn from prior IAA and MLLM-based evaluation settings.

Metric Category	Representative Metrics	Score Reg.	Text-Aug.	MLLM Reasoning
Statistical Corr.	PLCC, SRCC, KROCC	✓	×	✓
	MAE, MSE, RMSE, EMD	✓	×	✓
Lexical Overlap (NLG)	BLEU, ROUGE	×	✓	✓
	METEOR	×	✓	✓
Semantic Reasoning	BERTScore, SBERT	×	✓	✓
	VQA Acc.,	×	×	✓
	LLM-as-a-Judge			

The historical trajectory of these datasets reveals a fundamental evolution in how the computer vision community conceptualizes beauty: a gradual shift from treating aesthetics as a

scalar label to understanding it as a structured, explainable, and language-mediated concept. Early benchmarks reduced the complex cognitive process of aesthetic appreciation to a binary signal, treating the assessment model as a black box discriminator. However, the successive introduction of attributes, linguistic critiques, and psychological priors demonstrates a clear trend towards interpretability. Crucially, these datasets should not be viewed merely as alternative benchmarks for a single task but rather as definitions of fundamentally distinct aesthetic problems. The choice of a dataset dictates the mathematical formulation of the objective function: AVA defines aesthetics as generic score regression, whereas PARA redefines it as personalized ranking conditioned on user profiles. Fig. 5 illustrates representative data formats used in IAA benchmarks, including numerical ratings, VQA-style aesthetic questions, and language-based reasoning annotations. Consequently, the diversity of current datasets implies that IAA has evolved into a composite field, comprising a spectrum of sub-tasks ranging from objective quality quantification to subjective semantic interpretation.

Evaluation Metrics

As IAA has expanded from scalar score prediction to language-based outputs, the associated evaluation metrics have also evolved from statistical correlation measures to text-oriented and semantic similarity metrics. Table 5 summarizes the applicability of different evaluation metrics across IAA paradigms.

For score prediction tasks in IAA, evaluation metrics are typically divided into correlation-based and error-based measures. Correlation-based metrics assess the consistency between predicted scores and ground-truth rankings, while error-based metrics quantify absolute differences in score values.

Correlation-based Metrics Correlation-based metrics evaluate the consistency between predicted scores and ground-truth rankings, focusing on relative ordering rather than absolute values.

Pearson Linear Correlation Coefficient (PLCC)^{14,77} measures the linear correlation between predicted scores and ground-truth values. It reflects linear agreement between predicted scores and human ratings. A higher PLCC indicates stronger linear agreement.

$$\text{PLCC} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}}, \quad (1)$$

where x_i and y_i denote the predicted and ground-truth aesthetic scores of the i -th sample, respectively, $\bar{x}$ and $\bar{y}$ are their mean values, and N is the total number of samples.

In some image quality assessment protocols, nonlinear regression mapping is applied before computing PLCC to reduce scale mismatch^{78,79}.

Spearman Rank Order Correlation Coefficient (SRCC)^{14,80} evaluates the monotonic relationship between predicted and ground-truth rankings, focusing on whether the relative ordering of images is preserved:

$$\text{SRCC} = 1 - \frac{6 \sum_{i=1}^N d_i^2}{N(N^2 - 1)}, \quad (2)$$

where d_i is the difference between the predicted rank and ground-truth rank of the i -th sample. This formulation assumes no tied ranks; otherwise, rank-based implementations are used in practice.

Kendall Rank Correlation Coefficient (KROCC)^{14,81} measures pairwise ranking consistency by comparing concordant and discordant pairs: 387
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$$\text{KROCC} = \frac{N_c - N_d}{\frac{1}{2}N(N-1)}, \quad (3)$$

where N_c and N_d denote the numbers of concordant and discordant pairs, respectively. Compared to SRCC, KROCC provides a stricter measure of ranking agreement. 389
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Error-based Metrics Error-based metrics measure numerical differences between predicted scores and human annotations, reflecting alignment with dataset-specific rating distributions rather than ranking consistency. 391
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Mean Absolute Error (MAE)^{71,82} computes the average absolute difference between predicted and ground-truth scores: 394
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$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |x_i - y_i| \quad (4)$$

Mean Squared Error (MSE) penalizes larger errors more heavily: 396

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2 \quad (5)$$

Root Mean Squared Error (RMSE) is the square root of MSE and preserves the original score scale: 397
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$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2} \quad (6)$$

In contrast, generative aesthetic assessment involves evaluating language-based outputs, where the quality of generated descriptions and their semantic alignment with visual content become central. We adopt natural language generation (NLG) metrics^{61,62} to measure the quality of generated aesthetic descriptions. These metrics can be broadly categorized into lexical-overlap metrics, semantic similarity metrics, and task-level evaluation metrics. However, these metrics primarily capture surface-level similarity or semantic closeness, and do not directly measure the correctness or faithfulness of aesthetic reasoning. 399
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Lexical-Overlap Metrics Lexical-overlap metrics evaluate surface-level similarity between generated descriptions and reference annotations based on n-gram matching. 406
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Bilingual Evaluation Understudy (BLEU)^{62,83} measures n-gram precision with a brevity penalty to discourage overly short outputs: 408
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$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right), \quad (7)$$

where p_n is the modified precision for n-grams of length n , w_n are weighting coefficients (typically uniform), and BP is the brevity penalty. In practice, BLEU-1 to BLEU-4 correspond to $n = 1$ to 4, capturing progressively longer n-gram dependencies. However, BLEU is sensitive to exact word matching and often fails to reflect semantic equivalence between different valid descriptions. 410
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ROUGE^{62,84} evaluates recall-oriented overlap between generated and reference texts. Common variants include ROUGE-1 and ROUGE-2 (unigram and bigram overlap), and ROUGE-L, which measures the longest common subsequence (LCS) between sentences.

METEOR^{61,85} improves upon BLEU by incorporating stemming, synonym matching, and flexible alignment, resulting in better correlation with human judgments.

Despite these improvements, lexical-overlap metrics remain fundamentally limited for aesthetic assessment, as multiple valid descriptions may differ significantly in wording while conveying similar aesthetic judgments.

Semantic Similarity Metrics To address the limitations of lexical matching, semantic similarity metrics evaluate deeper alignment between generated and reference texts.

BERTScore^{29,86} computes token-level semantic similarity using contextual embeddings from pretrained language models. Precision and recall are defined based on cosine similarity between token embeddings, and combined as:

$$\text{BERTScore-F1} = \frac{2 \cdot P \cdot R}{P + R}, \quad (8)$$

where P and R denote precision and recall computed over token-level embedding similarities.

SBERT-based similarity²⁹ measures sentence-level semantic alignment by computing cosine similarity between sentence embeddings:

$$\text{Cosine}(u, v) = \frac{u \cdot v}{\|u\| \|v\|}, \quad (9)$$

where u and v are the sentence embeddings of the generated and reference texts, respectively.

Cross-modal Semantic Similarity Metric For multimodal aesthetic evaluation, it is also important to measure the alignment between generated textual descriptions and input images. CLIPScore⁸⁷ leverages pretrained vision-language models to compute image-text similarity:

$$\text{CLIPScore}(I, T) = \frac{f_I(I) \cdot f_T(T)}{\|f_I(I)\| \|f_T(T)\|}, \quad (10)$$

where $f_I(\cdot)$ and $f_T(\cdot)$ denote the image and text encoders of CLIP, respectively. CLIPScore reflects cross-modal semantic alignment between visual content and generated text.

Overall, while semantic similarity metrics provide a stronger proxy than lexical metrics, they remain limited in evaluating aesthetic reasoning, as they rely on similarity to reference texts or embeddings rather than explicit assessment of judgment correctness and explanatory coherence.

Discussion Evaluation in image aesthetic assessment increasingly requires a dual perspective that considers both score prediction and language-based outputs. Correlation- and error-based metrics capture numerical consistency with human ratings, while generation-oriented metrics assess the quality and alignment of aesthetic explanations. These metrics are complementary, reflecting both quantitative agreement and interpretability.

However, existing metrics remain limited in capturing the subjective and expert-driven nature of aesthetics. Lexical and semantic similarity primarily measure textual alignment, but fail to explicitly reflect aesthetic principles such as composition or visual harmony. Likewise, strong score correlation does not necessarily indicate true aesthetic understanding.

Recent efforts explore large language models as evaluators (LLM-as-a-judge)⁸, enabling more flexible and human-aligned assessment. However, this approach introduces new challenges, including evaluation instability, prompt sensitivity, and potential bias inherited from pre-trained models. Moreover, the lack of standardized protocols makes results difficult to reproduce and compare across studies. Future work should focus on developing aesthetic-aware evaluation protocols that explicitly incorporate domain knowledge, visual grounding, and reasoning consistency, bridging the gap between quantitative metrics and human aesthetic judgment.

Experiments and Benchmark

This section benchmarks representative IAA models under two complementary settings: numerical score prediction and language-based aesthetic assessment. For score prediction, we compare MLLMs with score-centric vision models using correlation-based and error-based metrics. For language-based evaluation, we use lexical-overlap and semantic similarity metrics for generated descriptions, together with accuracy on the VQA-style perception subset of UNIAA-Bench. By combining score prediction, description generation, and perception-oriented evaluation, our benchmark avoids relying on a single leaderboard metric and follows the broader view that benchmark suites can provide more informative model assessment than single-task rankings⁸⁸.

Experimental Setup

Following the evaluation framework introduced above, we conduct experiments from two complementary perspectives: (1) score prediction and (2) language-based evaluation.

Tasks and Datasets For score prediction, we adopt two widely used benchmarks: AVA⁷¹ and TAD66K⁸⁹, following their official test splits. AVA contains 19,930 test images with aesthetic ratings, and TAD66K contains 14,079 images with scores ranging from 1 to 10. For language-based evaluation, we utilize the UNIAA-Bench dataset²⁹, including the *description* subset for generated aesthetic descriptions and the *perception* subset for VQA-style aesthetic perception.

Comparison Methods To comprehensively reflect the evolution of IAA, we group the comparison methods into three categories: (1) Score-centric vision models, including CNN-based architectures and vision-language models such as CLIP-based methods; (2) General MLLMs, including InternVL, Qwen-VL, and LLaVA series; and (3) IAA-specific MLLMs, which are fine-tuned or adapted for aesthetic assessment.

For score prediction, General MLLMs are evaluated with direct prompting to obtain numerical predictions, following common prompt-based interaction protocols for large language models⁹⁰. Specifically, each model is prompted with: “Rate the aesthetic quality of this image from 1 to 10. Only output a single float number.” We constrain the output format and extract the predicted floating-point value from each response. For IAA-specific MLLMs, we follow the official scoring protocols provided by the authors, so that each model is evaluated under its intended scoring setup. For language-based evaluation on UNIAA-Bench, all models use the official prompt provided by the benchmark for the corresponding task. Across all evaluations, we keep dataset splits, metrics, and result reporting consistent to improve comparability and reproducibility, following the broader emphasis on standardized benchmark platforms⁹¹.

Metrics and Visualization For score prediction, we use both correlation-based metrics PLCC, SRCC, KROCC, and error-based metrics, MAE, MSE, and RMSE^{14,71}. We treat correlation-based metrics as the primary indicators because they measure consistency with human preference rankings, while error-based metrics are reported as supplementary measures of alignment with dataset-specific human rating distributions.

For language-based evaluation on the *description* subset, we use lexical-overlap metrics, including ROUGE (ROUGE-1, ROUGE-2, ROUGE-L), BLEU, and METEOR, and semantic similarity metrics, including BERTScore (P/R/F1) and SBERT cosine similarity^{29,61,62}. Specifically, BERTScore is computed using *deberta-base-mnli*, and SBERT is computed using *all-mpnet-base-v2*. BLEU is computed at the corpus level using SacreBLEU. Following the SacreBLEU output convention, BLEU is reported on a 0–100 scale. For the *perception* subset, we report accuracy under a VQA-style setting⁹².

To make multi-metric results easier to visualize, we aggregate selected metrics into compact visualization scores. These aggregated scores are used only for figures and do not replace the original metrics reported in the tables.

For score prediction, PLCC, SRCC, and KROCC are all correlation-based metrics with the same direction, where higher values indicate better agreement with human preference rankings. Therefore, we average them to obtain a correlation-based visualization score:

$$\text{Score}_{\text{corr}} = \frac{1}{3} (\text{PLCC} + \text{SRCC} + \text{KROCC}). \quad (11)$$

For error-based metrics, lower MAE and RMSE indicate better performance, which is opposite to the direction of correlation-based metrics. To make the scatter plots easier to interpret, we first apply min–max normalization to MAE and RMSE separately within the set of models included in the same visualization and then reverse their direction so that higher values indicate lower prediction error:

$$\widehat{E} = \frac{E - \min(E)}{\max(E) - \min(E)}, \quad E \in \{\text{MAE}, \text{RMSE}\}, \quad (12)$$

$$\text{Score}_{\text{error}} = 1 - \frac{1}{2} (\widehat{\text{MAE}} + \widehat{\text{RMSE}}). \quad (13)$$

For figures that jointly summarize AVA and TAD66K, we first compute the visualization scores on each dataset and then average the dataset-level scores.

For language-based evaluation, BERTScore-F1 and SBERT cosine similarity are both semantic similarity metrics with the same direction. We average them to obtain a semantic similarity visualization score:

$$\text{Score}_{\text{sem}} = \frac{1}{2} (\text{BERT-F1} + \text{SBERT-Cos}). \quad (14)$$

For lexical-overlap metrics, ROUGE-L and METEOR are reported on a 0–1 scale, whereas BLEU is reported on a 0–100 scale following the SacreBLEU output convention. We therefore rescale BLEU before combining it with ROUGE-L and METEOR:

$$\text{BLEU}_{\text{norm}} = 0.1 + 0.2 \cdot \frac{\text{BLEU} - \min(\text{BLEU})}{\max(\text{BLEU}) - \min(\text{BLEU})}. \quad (15)$$

The lexical-overlap visualization score is then computed as:

$$\text{Score}_{\text{lex}} = \frac{1}{3} (\text{R-L} + \text{METEOR} + \text{BLEU}_{\text{norm}}). \quad (16)$$

IAA Score Prediction Evaluations and Benchmarks

This section analyzes model performance on AVA and TAD66K, focusing on how different model families perform under correlation-based and error-based metrics.

Table 6: Comparison of representative image aesthetic assessment (IAA) models on the AVA dataset using PLCC, SRCC, KROCC, MAE, MSE, and RMSE metrics. InternVL denotes InternVL3.5-8B, QwenVL denotes Qwen3VL-8B, and LLaVA denotes LLaVA-OneVision-1.5-8B-Instruct. Higher values are better for PLCC, SRCC, and KROCC, while lower values are better for MAE, MSE, and RMSE. The comparison indicates that score-centric vision models achieve the strongest overall performance in numerical score prediction, while General MLLMs show weaker score-level alignment. The optimal result in each group is highlighted in bold.

Model	PLCC↑	SRCC↑	KROCC↑	MAE↓	MSE↓	RMSE↓
Score-centric vision models						
MUSIQ (2021) ¹⁵	0.8236	0.8225	0.6282	0.3033	0.1827	0.4274
EAT (2023) ⁹³	0.8755	0.8763	0.6925	1.0823	1.5809	1.2574
TANet (2022) ⁹⁴	0.7885	0.7891	0.5909	0.6013	0.5258	0.7251
VILA (2023) ²¹	0.7747	0.7759	0.5785	0.6248	0.6148	0.7841
UniQA (2025) ²⁰	0.5286	0.5177	0.3612	0.7750	0.8589	0.9267
General MLLMs						
InternVL (2025) ⁹⁵	0.3400	0.3348	0.2585	2.4218	6.6856	2.5856
QwenVL (2025) ⁹⁶	0.4295	0.4107	0.3223	1.7546	3.7249	1.9300
LLaVA (2025) ⁹⁷	0.4413	0.4412	0.3391	2.4494	6.8886	2.6246
IAA-Specific MLLMs						
ArtiMuse (2025) ³¹	0.5159	0.5383	0.3791	0.7134	0.9085	0.9531
ArtQuant (2026) ⁶⁵	0.4017	0.3911	0.2813	1.1645	2.0839	1.4436
OneAlign (2024) ⁹⁸	0.8187	0.8224	0.6295	0.9069	1.2552	1.1203
Q-Sit (2025) ⁹⁹	0.3277	0.2793	0.2219	3.0012	10.7399	3.2772
UNIAA (2024) ²⁹	0.7070	0.7131	0.5568	1.3258	1.5224	1.2338
AesExpert (2024) ³⁰	0.3421	0.3258	0.3452	1.8123	2.5417	1.5943

Quantitative Evaluations on Score Prediction Performance

Tables 6 and Tables 7 report PLCC, SRCC, KROCC, MAE, MSE, and RMSE for different methods on AVA and TAD66K. score-centric vision models achieve the strongest overall performance in numerical aesthetic score prediction, especially under correlation-based metrics. Fig. 6 provides a compact view of this pattern: General MLLMs are mainly located in the lower-left region, whereas strong score-centric vision models move toward the upper-right region. Several IAA-specific MLLMs approach this region on some metrics, but MLLM-based methods do not consistently surpass score-centric vision models on AVA and TAD66K. This indicates that general multimodal pretraining, language generation, and reasoning ability alone do not directly translate into stronger aesthetic score regression.

The performance differences within IAA-oriented models further indicate that aesthetic-specific training is beneficial but not uniformly effective across datasets. For example, EAT achieves the strongest correlation-based performance on AVA, while MUSIQ obtains lower error-based metrics, showing that ranking consistency and score distribution alignment are not always optimized by the same model. On TAD66K, UNIAA and OneAlign remain competitive with strong score-centric vision models, but some IAA-specific models show substantially lower performance. This variation suggests that IAA models can still be affected by dataset-specific rating distributions

Table 7: Comparison of representative image aesthetic assessment (IAA) models on the TAD66K dataset using the same metrics as Table 6. Compared with AVA, most models show lower correlation-based performance, while OneAlign and UNIAA still achieve strong performance on TAD66K.

Model	PLCC↑	SRCC↑	KROCC↑	MAE↓	MSE↓	RMSE↓
Score-centric vision models						
MUSIQ (2021)	0.4362	0.4085	0.2802	1.1032	1.7242	1.3131
EAT (2023)	0.4461	0.4168	0.2857	1.4431	3.1712	1.7808
TANet (2022)	0.4379	0.4131	0.2828	1.1939	2.0657	1.4373
VILA (2023)	0.4433	0.4182	0.2869	1.1348	1.9109	1.3824
UniQA (2025)	0.3607	0.3324	0.2259	1.2221	2.2240	1.4913
General MLLMs						
InternVL (2025)	0.2202	0.2050	0.1581	2.7953	9.8803	3.1432
QwenVL (2025)	0.2522	0.2268	0.1741	2.0762	6.0426	2.4582
LLaVA (2025)	0.2836	0.2641	0.2032	2.9341	10.6879	3.2695
IAA-Specific MLLMs						
ArtiMuse (2025)	0.2454	0.2478	0.1673	1.2849	2.5156	1.5861
ArtQuant (2026)	0.2636	0.2478	0.1664	1.8423	5.0460	2.2463
OneAlign (2024)	0.4451	0.4170	0.2861	1.2229	2.2448	1.4983
Q-Sit (2025)	0.2995	0.2859	0.1923	3.1683	12.4004	3.5214
UNIAA (2024)	0.4112	0.4203	0.3012	1.5583	2.8133	1.6773
AesExpert (2024)	0.1966	0.2113	0.1653	4.6324	23.434	4.8409

and annotation preferences. Compared with AVA, TAD66K generally leads to lower scores under the same metric groups, suggesting that it poses a stronger challenge for score prediction under our evaluation setting.

Table 8: Performance of InternVL and Qwen3-VL models with different parameter scales on the AVA dataset. The results show that larger models generally improve correlation-based performance, while error-based metrics do not always improve monotonically.

Model	PLCC↑	SRCC↑	KROCC↑	MAE↓	MSE↓	RMSE↓
InternVL Series						
InternVL-38B	0.4204	0.4186	0.3302	2.5955	7.4560	2.7306
InternVL-8B	0.3400	0.3348	0.2585	2.4218	6.6856	2.5856
InternVL-4B	0.3322	0.3346	0.2699	2.5911	7.3613	2.7132
InternVL-2B	0.2132	0.1776	0.1415	2.2889	8.7599	2.9597
Qwen3-VL Series						
Qwen3VL-30B	0.4737	0.4630	0.3449	1.4374	2.7692	1.6641
Qwen3VL-8B	0.4295	0.4107	0.3223	1.7546	3.7249	1.9300
Qwen3VL-4B	0.4225	0.4107	0.3223	1.9642	4.4788	2.1161
Qwen3VL-2B	0.3575	0.3436	0.2559	0.8388	1.0951	1.0465

Tables 8 and 9 further examine the effect of model scale. The results show that larger General MLLMs usually achieve stronger correlation-based performance, especially in the Qwen3-VL series, where the 30B model obtains the best PLCC, SRCC, and KROCC on both AVA and TAD66K. Fig. 7 makes this trend clearer by showing that PLCC and SRCC generally increase with model scale, whereas MAE and MSE fluctuate more strongly across models. This indicates that model scale is more closely related to ranking consistency than to score distribution alignment. Smaller models sometimes obtain lower MAE or RMSE, but this does not necessarily indicate stronger aesthetic understanding. A likely explanation is that smaller models produce

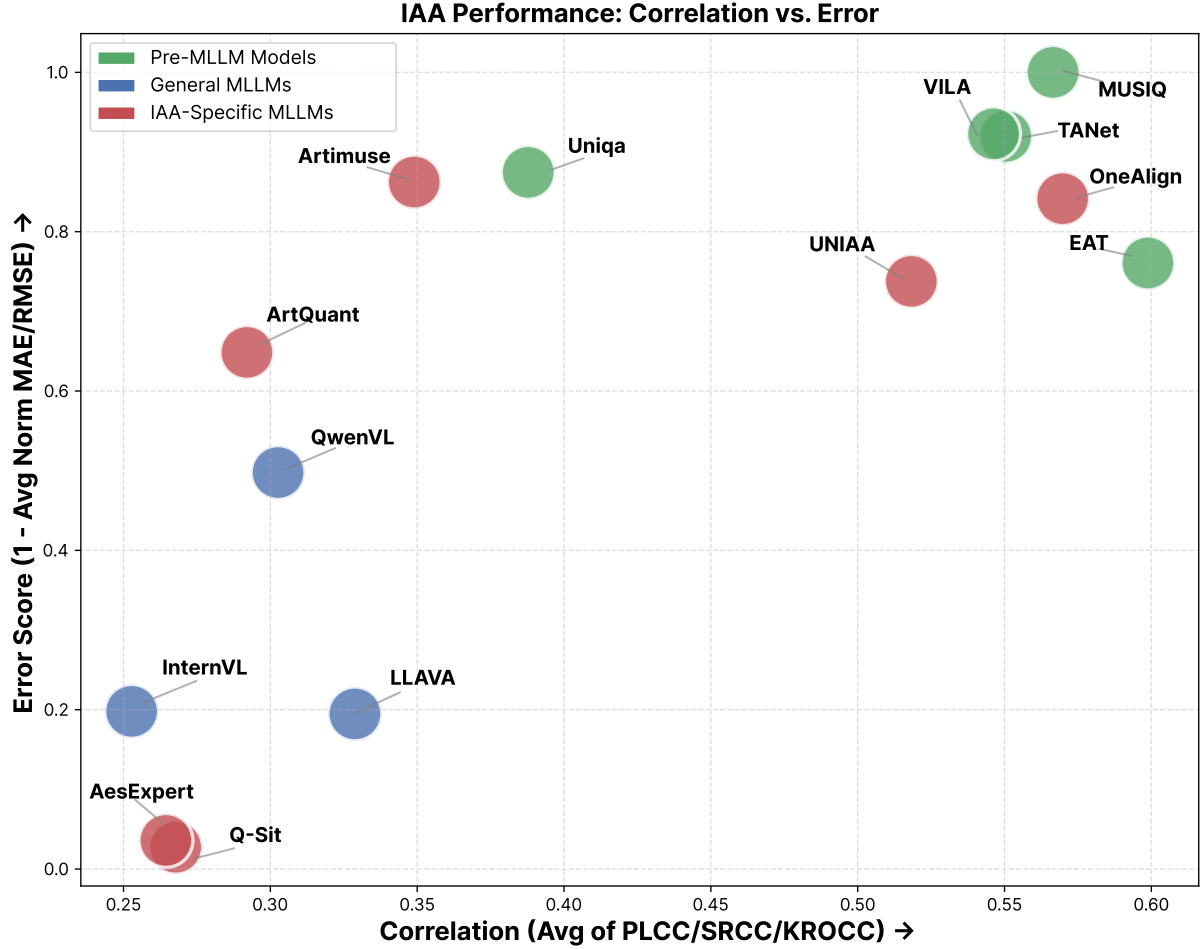


Figure 6: Visualization of score prediction performance using aggregated correlation-based and error-based scores. The x-axis reports the mean of PLCC, SRCC, and KROCC, and the y-axis reports the reversed normalized error-based score computed from MAE and RMSE. Higher values on both axes indicate better performance. The aggregated scores are used only for visualization.

Table 9: Performance of InternVL and Qwen3-VL models with different parameter scales on the TAD66K dataset. The results further show that model scale is more consistently associated with ranking correlation than with numerical error.

Model	PLCC↑	SRCC↑	KROCC↑	MAE↓	MSE↓	RMSE↓
InternVL Series						
InternVL-38B	0.2471	0.2425	0.1904	2.9532	10.6475	3.2631
InternVL-8B	0.2202	0.2050	0.1581	2.7953	9.8803	3.1432
InternVL-4B	0.1927	0.1853	0.1502	2.818	9.9536	3.1549
InternVL-2B	0.1053	0.0879	0.0711	2.9989	11.0360	3.3306
Qwen3-VL Series						
Qwen3VL-30B	0.2964	0.2775	0.2017	1.8969	5.1993	2.2802
Qwen3VL-8B	0.2522	0.2268	0.1741	2.0762	6.0426	2.4582
Qwen3VL-4B	0.2595	0.2421	0.1904	2.3203	7.2281	2.6885
Qwen3VL-2B	0.1693	0.1442	0.1105	1.3409	2.7356	1.6540

more conservative score distributions, which can reduce numerical error without improving preference ranking. Overall, model size can improve score prediction when the scale increase is substantial, but scaling alone is not sufficient to close the gap with score-centric vision models. This suggests that IAA score prediction depends not only on general multimodal capacity, but also on task-specific supervision and numerical alignment.

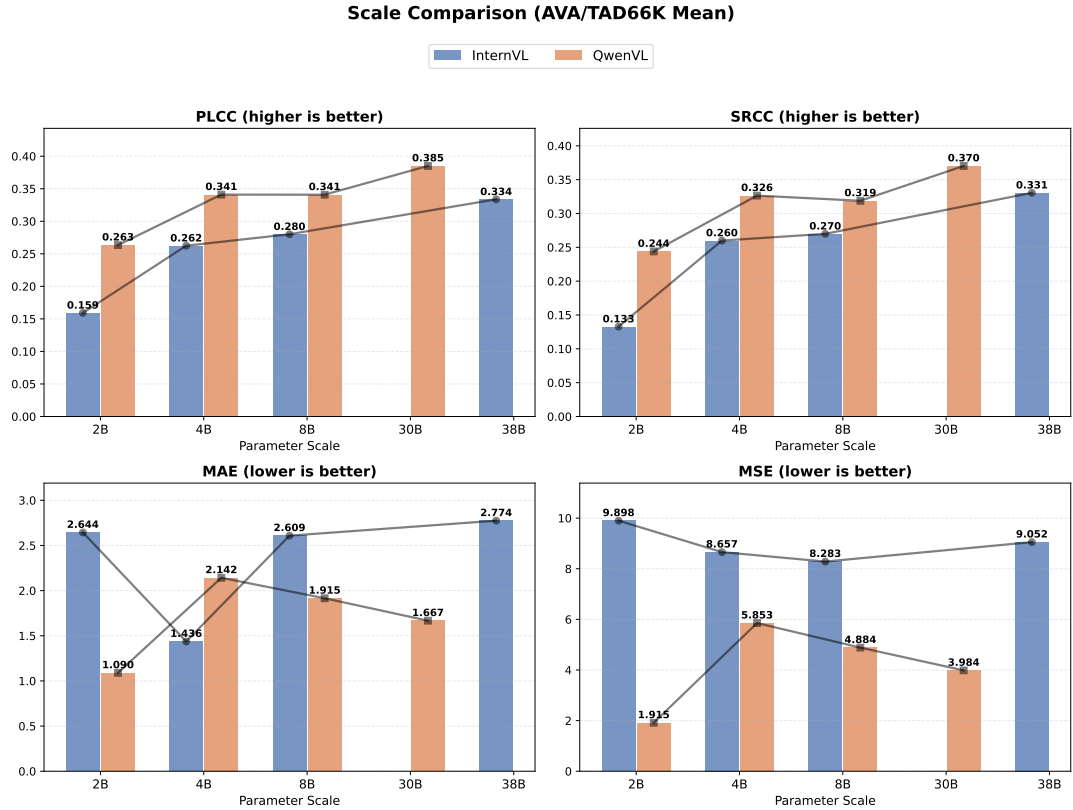


Figure 7: Effect of model scale on score prediction performance. The figure reports averaged results on AVA and TAD66K for the InternVL and Qwen3-VL series. PLCC and SRCC are higher-is-better metrics, while MAE and MSE are lower-is-better metrics.

Because several compared models provide versions fine-tuned on different datasets, we further evaluate their cross-dataset generalization. The results are shown in Table 10, together with Fig. 8. On the AVA test set, ArtiMuse-AVA and EAT-AVA achieve higher PLCC, SRCC, and KROCC than the corresponding models trained on TAD66K, indicating stronger ranking consistency when training and evaluation are performed on the same benchmark. On the TAD66K test set, ArtiMuse-TAD66K and EAT-TAD66K also achieve higher correlation-based performance than their AVA-trained counterparts, with the same trend observed in error-based metrics.

These cross-dataset results show that dataset-specific fine-tuning improves in-domain ranking consistency but does not guarantee stable generalization across benchmarks. Error-based metrics provide a supplementary view: for example, EAT-TAD66K obtains lower error-based metrics than EAT-AVA on the AVA test set, despite lower PLCC, SRCC, and KROCC. This indicates that a model may better match the rating distribution of a dataset while showing weaker consistency with human preference rankings. Overall, score prediction in IAA should be evaluated with correlation-based metrics as the primary evidence and error-based metrics as complementary measures. The results also show that current IAA models remain sensitive to dataset-specific rating distributions and annotation preferences, highlighting the need for evaluation protocols beyond in-domain testing.

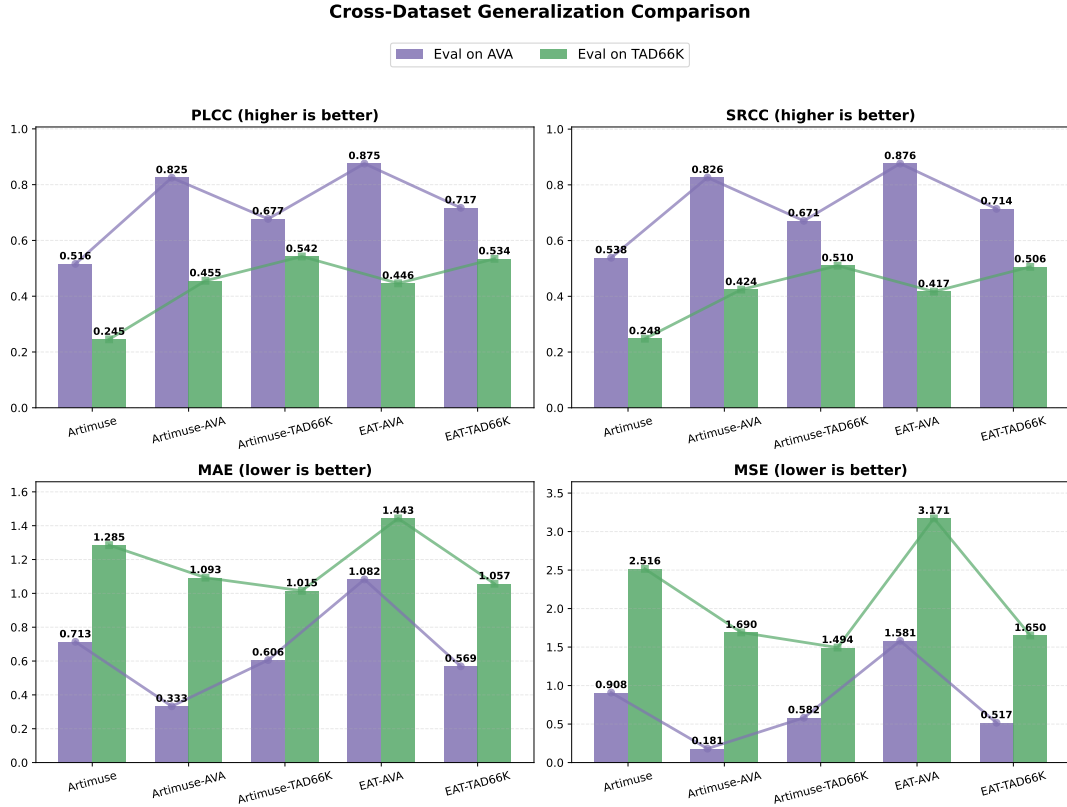


Figure 8: Cross-dataset evaluation of ArtIMuse and EAT with dataset-specific fine-tuned variants on AVA and TAD66K. The grouped bar charts show that in-domain fine-tuning improves performance, while cross-dataset transfer remains limited.

Disadvantage Analysis of MLLMs in Score Prediction

As shown in Tables 6 and 7, MLLM-based methods, including General MLLMs and IAA-specific MLLMs, do not consistently surpass score-centric vision models under correlation-based metrics. Although MLLMs usually have larger model capacity and stronger multimodal semantic modeling, these advantages are not directly reflected in score prediction on AVA and TAD66K. This gap suggests that language-oriented MLLMs still face a mismatch when applied to aesthetic score regression. We discuss three possible reasons below.

First, score-centric vision models are usually trained on aesthetic rating datasets with losses that directly target score prediction. This training setup makes them better suited to matching human preference rankings on benchmarks such as AVA and TAD66K. By contrast, MLLM-based models are often optimized for language generation, instruction following, or joint scoring and explanation, rather than direct regression to continuous aesthetic scores. As a result, a model may generate informative aesthetic descriptions while still showing weaker correlation with human rating distributions.

Second, there is a gap between token-based generation and continuous score prediction^{31,69}. MLLMs generate discrete tokens, whereas aesthetic scores are continuous values. When scores are obtained through prompting, discrete labels, or token-to-score mapping, the resulting predictions can be less stable than outputs from dedicated regression heads. This makes it difficult for MLLM-based methods to achieve the same level of ranking consistency as score-centric vision models.

Third, score prediction benchmarks measure agreement with dataset-specific human rating distributions rather than aesthetic understanding in a broader sense. High PLCC or SRCC may partly reflect how well a model matches the annotation pattern of a given dataset. Therefore, lower correlation scores for MLLM-based methods should not be taken as direct evidence of

Table 10: Cross-dataset evaluation results of the ArtiMuse and EAT series on AVA and TAD66K. Models fine-tuned on different source datasets are compared under the same evaluation protocol. The results indicate that matching the training and evaluation datasets improves correlation-based performance, suggesting that current IAA models remain sensitive to dataset-specific rating distributions.

A. Evaluation on the AVA test set						
Model	PLCC↑	SRCC↑	KROCC↑	MAE↓	MSE↓	RMSE↓
ArtiMuse Series						
ArtiMuse	0.5159	0.5383	0.3791	0.7134	0.9085	0.9531
ArtiMuse-AVA	0.8249	0.8263	0.6326	0.3326	0.1808	0.4252
ArtiMuse-TAD66K	0.6771	0.6709	0.4833	0.6057	0.5819	0.7628
EAT Series						
EAT-AVA	0.8755	0.8763	0.6925	1.0823	1.5809	1.2574
EAT-TAD66K	0.7169	0.7138	0.5211	0.5689	0.5170	0.7190

B. Evaluation on the TAD66K test set						
Model	PLCC↑	SRCC↑	KROCC↑	MAE↓	MSE↓	RMSE↓
ArtiMuse Series						
ArtiMuse	0.2454	0.2478	0.1673	1.2849	2.5156	1.5861
ArtiMuse-AVA	0.4553	0.4240	0.2912	1.0928	1.6900	1.2998
ArtiMuse-TAD66K	0.5425	0.5099	0.3555	1.0154	1.4935	1.2221
EAT Series						
EAT-AVA	0.4461	0.4168	0.2857	1.4431	3.1712	1.7808
EAT-TAD66K	0.5340	0.5057	0.3518	1.0568	1.6503	1.2846

weaker aesthetic understanding. Instead, they show that current MLLMs are not yet fully aligned with score prediction metrics, while their strengths may be better reflected in language-based evaluation and fine-grained explanation.

These factors help explain why MLLM-based methods still lag behind score-centric vision models in numerical score prediction, despite their stronger general multimodal capabilities. Future IAA models may need stronger links between language-based aesthetic analysis and score prediction, such as score-aware prompting, token-to-score alignment, fine-grained attribute supervision, or lightweight regression heads.

IAA Reasoning Evaluations and Benchmarks

This section evaluates MLLMs on UNIAA-Bench under language-based evaluation. We consider two complementary subsets: the *description* subset, which evaluates generated aesthetic analyses using lexical-overlap and semantic similarity metrics, and the *perception* subset, which evaluates VQA-style aesthetic perception using accuracy. Tables 11 and 12, together with Figs. 9, 10, 11, and 12, provide complementary views of model behavior across these evaluation settings.

Lexical and Semantic Evaluations on Description Subset

For the description subset, Tables 11 and 12 report lexical-overlap and semantic similarity scores for generated aesthetic descriptions. Fig. 9 summarizes the lexical-semantic distribution using aggregated visualization scores, while Fig. 10 breaks down the performance of selected representative models across individual metrics.

Under lexical-overlap metrics, General MLLMs remain competitive with IAA-specific MLLMs. LLaVA-OneVision-8B achieves scores close to the best IAA-specific MLLMs on R-1, R-L, and BLEU, while InternVL3.5-8B obtains the highest METEOR score. The lexical-overlap radar chart in Fig. 10 shows that selected General MLLMs and IAA-specific MLLMs occupy similar ranges on several lexical axes. This suggests that lexical matching captures surface similarity to reference descriptions, but does not clearly distinguish aesthetic-specific reasoning ability.

Table 11: Lexical-overlap metrics of MLLMs on the UNIAA-Bench description subset. The results show that General MLLMs remain competitive with IAA-specific MLLMs under surface-level text matching metrics, suggesting that lexical overlap alone is insufficient for judging aesthetic analysis quality.

Model	R-1 ↑	R-2 ↑	R-L ↑	METEOR ↑	BLEU ↑
General MLLMs					
Qwen3VL-8B	0.3232	0.0485	0.1776	0.1470	1.8529
InternVL3.5-8B	0.3391	0.0789	0.1837	0.2489	1.8064
LLaVA-OneVision-8B	0.3687	0.0831	0.2046	0.1981	3.8436
IAA-Specific MLLMs					
UNIAA	0.3293	0.0787	0.1976	0.2343	2.9932
AesExpert	0.3155	0.0851	0.2072	0.1233	2.3475
Q-SIT	0.2942	0.0697	0.1968	0.1159	1.4403
ArtiMuse	0.3604	0.0787	0.2030	0.1668	3.8659
ArtQuant	0.3688	0.0750	0.2146	0.1951	3.2621

Table 12: Semantic similarity metrics of MLLMs on the UNIAA-Bench description subset. The results show that the strongest IAA-specific MLLMs achieve better semantic alignment with reference descriptions, while performance varies substantially across models and metrics.

Model	BERT-P ↑	BERT-R ↑	BERT-F1 ↑	SBERT-Cos ↑
General MLLMs				
Qwen3VL-8B	0.5819	0.5990	0.5902	0.7058
InternVL3.5-8B	0.5981	0.6674	0.6308	0.7197
LLaVA-OneVision-8B	0.6483	0.6593	0.6533	0.7250
IAA-Specific MLLMs				
UNIAA	0.6365	0.6797	0.6572	0.7525
AesExpert	0.6898	0.6126	0.6487	0.6432
Q-SIT	0.6677	0.6017	0.6329	0.6422
ArtiMuse	0.6681	0.6554	0.6616	0.7382
ArtQuant	0.6457	0.6442	0.6449	0.6447

This limitation is further illustrated by the qualitative example in Fig. 11. Given the same image and prompt, different models produce descriptions with different levels of visual detail, compositional analysis, and explanatory focus. Some fluent responses remain relatively generic, while more focused outputs directly refer to composition, visual evidence, and aesthetic criteria. This suggests that lexical-overlap metrics are useful for measuring surface matching, but they are insufficient for assessing the precision and usefulness of aesthetic reasoning.

Under semantic similarity metrics, the strongest IAA-specific MLLMs show clearer advantages. ArtiMuse obtains the highest BERT-F1, and UNIAA achieves the highest SBERT-Cos score. The semantic radar chart in Fig. 10 shows that selected IAA-specific models achieve stronger semantic similarity than General MLLMs on several semantic metrics. This indicates that aesthetic-specific training can improve semantic alignment with reference aesthetic descriptions for some models, although the effect is not uniform across all IAA-specific MLLMs.

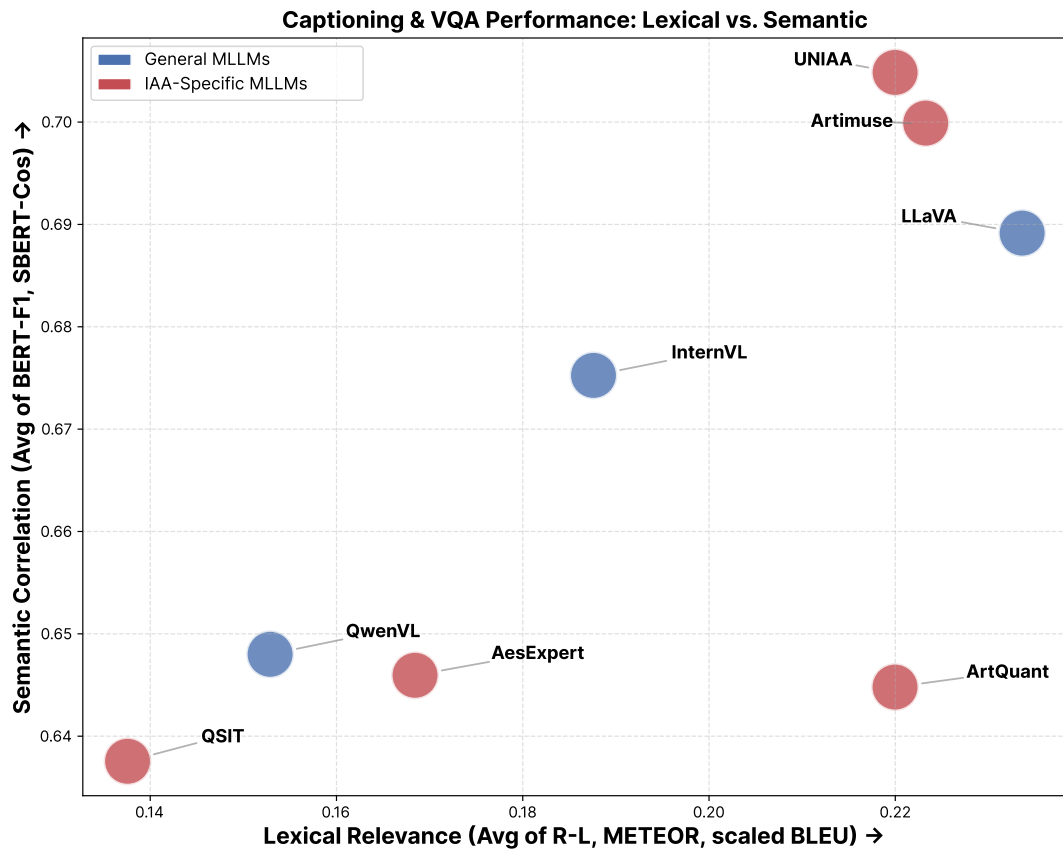


Figure 9: MLLMs comparison on UNIAA-Bench using aggregated lexical-overlap and semantic similarity scores. Higher values indicate stronger agreement with reference aesthetic descriptions under the corresponding metric group.

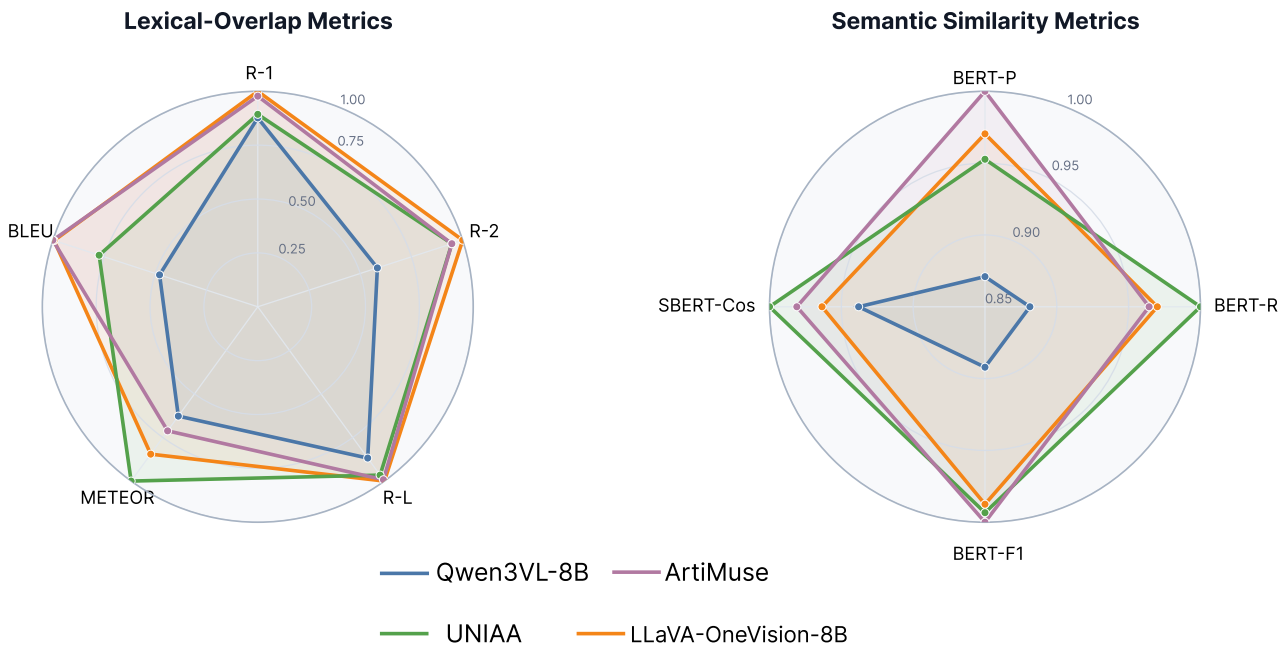


Figure 10: Radar visualization of MLLMs on UNIAA-Bench. The left chart compares lexical-overlap metrics, including R-1, R-2, R-L, METEOR, and BLEU, while the right chart compares semantic similarity metrics, including BERT-P, BERT-R, BERT-F1, and SBERT-Cos. Values are min-max normalized per metric for visualization.

Prompt: Describe and evaluate the aesthetic attributes in detail.



Input Image

UNIAA:

The image showcases a **well-composed travel photograph** of the city of Budapest, Hungary. The use of the camera is commendable, with a good depth of field and focus on the subject. The color and lighting are well-executed, with a nice balance **between the natural light and the artificial lighting**. The composition is visually appealing, with the **arches framing the cityscape** and **creating a sense of depth**. Overall, the image captures the beauty of the city and its architectural landmarks.

ArtQuant:

The image presents a captivating **view through a window**, showcasing a cityscape with **a prominent castle-like structure**. **The arched opening frames the scene**, drawing the viewer's attention to the architectural marvels beyond. **The cityscape is bustling with activity**, with numerous buildings and structures that create a sense of depth and scale. The color palette is **soft and muted**, evoking a **serene atmosphere**. The gentle hues of the sky and the water reflect a tranquil mood, while the intricate details of the buildings and the surrounding landscape invite closer inspection. The arched opening itself adds a touch of elegance, enhancing the overall aesthetic appeal of the scene. In summary, this image successfully combines elements of architecture, cityscape, and aesthetics, creating a harmonious visual experience that invites contemplation and appreciation of the beauty found in urban environments.

Artimuse:

The image presents a striking view of a grand, historic building **framed by an archway**, creating a sense of **depth and perspective**. The architectural details of the building, with its **ornate spires and domes**, are highlighted against a soft, pastel sky, enhancing the visual impact. **The composition effectively uses the arch to draw the viewer's eye** towards the central structure, emphasizing its grandeur. The muted color palette and the serene atmosphere contribute to a harmonious and timeless aesthetic. While the image is visually appealing, it could benefit from more dynamic lighting to accentuate textures and shadows, adding further depth and contrast. Overall, the photograph captures a serene and majestic scene with strong compositional elements.

LLaVA-OneVision:

The image showcases a beautiful architectural scene with the **Hungarian Parliament Building as the focal point**. The building's intricate design, featuring multiple spires and domes, is highlighted by the **soft lighting of the overcast sky**. The river adds a serene element to the composition, reflecting parts of the cityscape. The arch framing the view creates a **natural vignette effect**, drawing attention to the central structure. Overall, the image has a harmonious balance between the man-made structures and the natural elements, creating an aesthetically pleasing and tranquil atmosphere.

Qwen-VL:

This image presents a compelling and atmospheric urban vista, **framed through a dark, arched architectural opening** that acts as a **natural vignette**. The aesthetic is defined by a harmonious interplay of composition, color, and mood, **creating a sense of both grandeur and quiet contemplation**. Composition and Framing: The most striking aesthetic element is **the use of the arched window as a framing device**. This creates a strong **sense of depth and perspective**, drawing the viewer's eye directly to the central subject: the Hungarian Parliament Building. The dark, textured edges of the arch contrast sharply with the lighter scene beyond, enhancing the focus and lending a sense of looking out from a hidden or elevated vantage point. The symmetry of the arch and the building's own symmetrical architecture create a balanced, **almost classical composition**. The river below acts as a natural dividing line, grounding the scene and adding to the sense of scale. Color Palette and Tone: The image is rendered in a muted, desaturated palette, dominated by soft grays, pale blues, and off-whites. This is characteristic of a cool, overcast day, which lends a somber, contemplative mood to the scene. The sky is a gradient of pale gray and soft pinkish-orange near the horizon, introducing a subtle warmth that prevents the image from feeling cold or sterile. This restrained use of color contributes to a timeless, **almost cinematic quality**, emphasizing atmosphere over vibrancy.

Figure 11: Qualitative comparison of language-based aesthetic evaluation. Given the same image and prompt, different MLLMs produce aesthetic descriptions with different levels of visual detail, compositional analysis, and explanatory focus. Blue highlights indicate statements that directly capture relevant aesthetic factors or visual evidence, while orange highlights indicate relatively generic or less visually grounded descriptions.

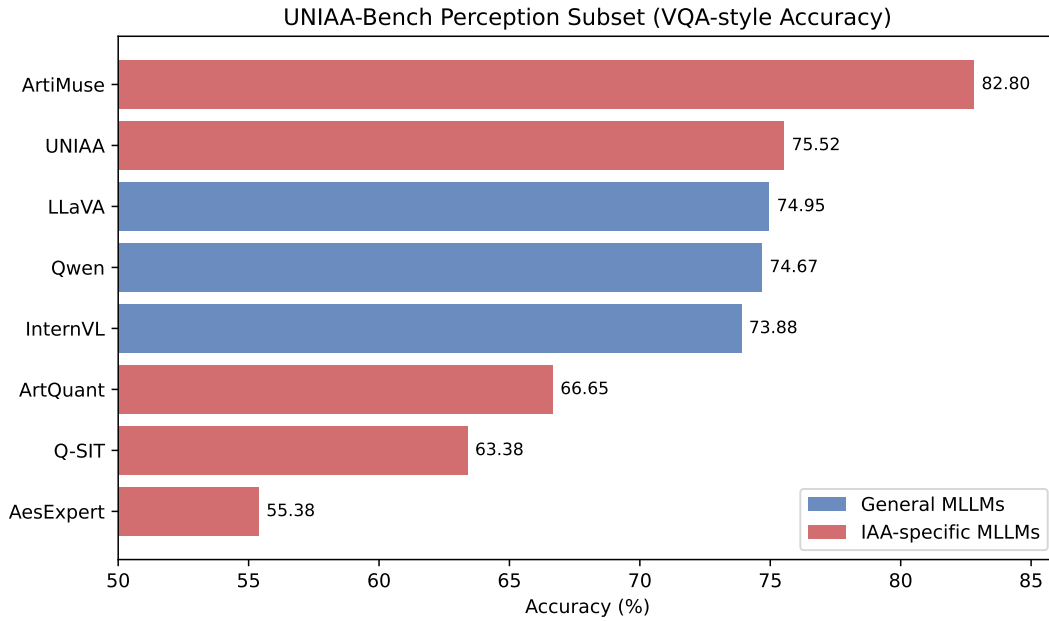


Figure 12: Accuracy comparison on the UNIAA-Bench perception subset under a VQA-style setting. ArtiMuse achieves the highest accuracy, while several IAA-specific MLLMs perform below General MLLMs, showing that aesthetic-specific training does not guarantee stronger perception performance across all models.

VQA Evaluations on Perception Subset

For the perception subset, models are evaluated under a VQA(Visual Question Answering)-style setting, where the task is to answer aesthetic perception questions rather than generate long-form descriptions. As shown in Fig. 12, ArtiMuse achieves the highest accuracy, while General MLLMs cluster around 74%. In contrast, several IAA-specific MLLMs, including AesExpert, Q-SIT, and ArtQuant, perform below General MLLMs. This result shows that aesthetic-specific training can improve perception accuracy when it is well aligned with the benchmark, but it does not guarantee stronger performance across all IAA-specific models.

Overall, the UNIAA-Bench results show that language-based aesthetic assessment cannot be summarized by a single metric. Lexical-overlap metrics mainly reflect surface text matching, semantic similarity metrics better capture reference-level semantic alignment, and VQA-style accuracy evaluates perception-oriented aesthetic understanding. General MLLMs show competitive surface-level text matching to IAA-specific MLLMs, whereas the strongest IAA-specific MLLMs show clearer advantages in semantic alignment and VQA-style perception. This finding suggests that future IAA benchmarks should evaluate not only whether a model produces fluent aesthetic descriptions, but also whether its explanations are visually grounded and aligned with aesthetic criteria.

Conclusion

This paper studies Image Aesthetic Assessment (IAA) through the lens of a scoring-to-reasoning transition. Rather than treating recent MLLM-based methods as a simple extension of previous scoring models, we organize IAA as an evolving task that moves from feature-based score prediction to vision-language alignment and further to reasoning-based aesthetic assessment. Based on this taxonomy, we analyze the evolution of datasets and evaluation protocols, and build a unified benchmark for comparing representative score-centric vision models, General MLLMs, and IAA-specific MLLMs on AVA, TAD66K, and UNIAA-Bench.

The experiments show that the current transition from scoring to reasoning is not a simple replacement of regression models by MLLMs. On AVA and TAD66K, score-centric vision models still achieve the strongest score prediction results, while MLLM-based methods show their main advantages in language-based aesthetic analysis. Larger model scale is generally associated with better ranking consistency, but it does not consistently improve error-based metrics, since lower numerical errors may also come from conservative score distributions. Cross-dataset results further show that current IAA models remain sensitive to dataset-specific rating distributions and annotation preferences. These findings indicate that score prediction benchmarks mainly evaluate agreement with dataset-specific rating distributions and human preference rankings, and therefore should not be treated as a complete measure of aesthetic understanding.

The results on UNIAA-Bench reveal another limitation: language-based evaluation cannot be reliably summarized by lexical-overlap metrics, semantic similarity metrics, or VQA-style accuracy alone. Aesthetic analysis is long-form and semantically open-ended, and a valid critique may use different wording or focus on different visual evidence from the reference description. Therefore, future IAA benchmarks should evaluate not only whether a model predicts a score or matches a reference sentence, but also whether its explanation identifies relevant aesthetic factors, follows coherent aesthetic criteria, and remains grounded in visual evidence. Taken together, our findings suggest that future IAA should connect reliable score prediction with visually grounded aesthetic reasoning, rather than treating scoring and explanation as separate outputs.

RESOURCE AVAILABILITY

Lead contact

Lead contact information will be added in the final submission.

Materials availability

This study did not generate new physical materials.

Data and code availability

Data and code availability statements will be added in the final submission.

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AUTHOR CONTRIBUTIONS

Author contributions will be added in the final submission.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

The declaration of generative AI and AI-assisted technologies will be added in the final submission.

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